

Key Findings:

01 Smoking

- Among people with hypertension: ~71.83% are smokers

02 Family History

- Hypertension group has a higher % with family history

03 BP History

- People with prior BP issues - much higher hypertension

04 Medication insight

- People with multiple medication categories are more likely to have hypertension. Patients with hypertension are more likely to be on antihypertensive medications, reflecting ongoing clinical management.

05 Exercise Level

Hypertension group:

- Higher % in Low exercise
- Lower % in High exercise

Overall Analysis Report

This analysis examines patient demographic, lifestyle, and medical history factors associated with hypertension. Using SAS, key patterns were identified to highlight individuals at higher risk and support earlier intervention.

Analytical Methods

Descriptive Statistics (Proc Means)	Categorical Analysis (Proc Freq)	Data Structure & Validation (Proc Contents)
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Provides an overview of key numerical variables by calculating averages and trends, helping identify differences in facts such as Age, BMI, Salt Intake, Stress Levels between groups.

Analyzes relationships between categorical variables by calculating frequencies and percentages, highlighting patterns such as smoking status, family history and exercise levels in relation to hypertension.

Examines the dataset structure, including variables names, types, and formats, ensuring data accuracy, consistency, and readiness for analysis.

Key Clinical Findings

The CONTENTS Procedure

Data Set Name	WORK.HYPER_CLEAN	Observations	1985
Member Type	DATA	Variables	11
Engine	V9	Indexes	0
Created	04/21/2026 11:59:36	Observation Length	96
Last Modified	04/21/2026 11:59:36	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	2
First Data Page	1
Max Obs per Page	1363
Obs in First Data Page	1329
Number of Data Set Repairs	0
Filename	/saswork/SAS_work715700002575_odaws01-usw2.oda.sas.com/SAS_work156900002575_odaws01-usw2.oda.sas.com/hyper_clean.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	3914630
Access Permission	rw-r--r--
Owner Name	u64128557
File Size	384KB
File Size (bytes)	393216

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
1	Age	Num	8	BEST12.	BEST32.
6	BMI	Num	8	BEST12.	BEST32.
4	BP_History	Char	15	\$15.	\$15.
9	Exercise_Level	Char	8	\$8.	\$8.
8	Family_History	Char	3	\$3.	\$3.
11	Has_Hypertension	Char	3	\$3.	\$3.
7	Medication	Char	13	\$13.	\$13.

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
2	Salt_Intake	Num	8	BEST12.	BEST32.
5	Sleep_Duration	Num	8	BEST12.	BEST32.
10	Smoking_Status	Char	10	\$10.	\$10.
3	Stress_Score	Num	8	BEST12.	BEST32.

The MEANS Procedure

Has_Hypertension	N Obs	Variable	Mean
No	953	Age	46.0797482
		BMI	25.3341028
		Salt_Intake	8.2942288
		Sleep_Duration	6.6440714
		Stress_Score	4.3683106
Yes	1032	Age	54.2761628
		BMI	26.6443798
		Salt_Intake	8.7509690
		Sleep_Duration	6.2750969
		Stress_Score	5.5436047

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of Has_Hypertension by Smoking_Status			
	Has_Hypertension	Smoking_Status		
		Non-Smoker	Smoker	Total
No		793	160	953
		39.95	8.06	48.01
		83.21	16.79	
		55.96	28.17	
Yes		624	408	1032
		31.44	20.55	51.99
		60.47	39.53	
		44.04	71.83	
Total		1417	568	1985
		71.39	28.61	100.00

Frequency Percent Row Pct Col Pct	Table of Has_Hypertension by Family_History			
	Has_Hypertension	Family_History		
		No	Yes	Total
No		620	333	953
		31.23	16.78	48.01
		65.06	34.94	
		62.00	33.81	
Yes		380	652	1032
		19.14	32.85	51.99
		36.82	63.18	
		38.00	66.19	
Total		1000	985	1985
		50.38	49.62	100.00

Frequency Percent Row Pct Col Pct	Table of Has_Hypertension by Exercise_Level				
	Has_Hypertension	Exercise_Level			
		High	Low	Moderate	Total
No		178	464	311	953
		8.97	23.38	15.67	48.01
		18.68	48.69	32.63	
		45.52	49.57	47.26	
Yes		213	472	347	1032
		10.73	23.78	17.48	51.99
		20.64	45.74	33.62	
		54.48	50.43	52.74	
Total		391	936	658	1985
		19.70	47.15	33.15	100.00

Frequency Percent Row Pct Col Pct	Table of Has_Hypertension by Medication						
	Has_Hypertension	Medication					Total
		ACE Inhibitor	Beta Blocker	Diuretic	None	Other	
No		162	198	106	387	100	953
		8.16	9.97	5.34	19.50	5.04	48.01
		17.00	20.78	11.12	40.61	10.49	
		50.15	48.06	43.27	48.44	48.54	
Yes		161	214	139	412	106	1032
		8.11	10.78	7.00	20.76	5.34	51.99
		15.60	20.74	13.47	39.92	10.27	
		49.85	51.94	56.73	51.56	51.46	
Total		323	412	245	799	206	1985
		16.27	20.76	12.34	40.25	10.38	100.00

Frequency Percent Row Pct Col Pct	Table of Has_Hypertension by BP_History				
	Has_Hypertension	BP_History			Total
		Hypertension	Normal	Prehypertension	
No		41	510	402	953
		2.07	25.69	20.25	48.01
		4.30	53.52	42.18	
		7.21	64.07	64.84	
Yes		528	286	218	1032
		26.60	14.41	10.98	51.99
		51.16	27.71	21.12	
		92.79	35.93	35.16	
Total		569	796	620	1985
		28.66	40.10	31.23	100.00

Conclusion & Clinical Insights

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Project: Hypertension Risk Analysis

Date: April 2026

Tools Used: SAS (Proc Means, Proc Freq, Proc Contents)

Conclusion

The analysis identified both modifiable and non-modifiable factors associated with hypertension. Patients with hypertension showed higher averages in age, BMI, salt intake and stress levels. In addition, lifestyle behaviors such as smoking and lower physical activity were more common among affected individuals.

These findings emphasize the importance of early detection, lifestyle intervention, and preventive care strategies to reduce hypertension risk and improve patients' outcomes.